

PART NUMBER	COMPONENT NAME	USE	Operational voltage	Current draw	POWER CONSUMPTION
HLK – 5M12	AC – DC converter	Power supply	12V	450mA – 1A	5 – 5.4W
L78M05ABDT - TR	5V LDO	Fixed 5v output	5V	5mA	25mW
LM1117DT -3.3	3.3V LDO	Fixed 3.3v output	3.3V	5mA	25mW
ESP32 WROOM 2	ESP32	controller	3.3V	10mA – 500mA	660mW
PC817	optocoupler	Circuit isolation	1.2V	5-20mA	6-24mW
BTA16 – 600B	triac	AC switching	1.5 – 3V	5 – 50mA	7.5 – 150mW
ACS712Xlctr-20A	Current sensor	AC current sensing	5v	10mA	50mW
TLV9001IDCK	Operational amp	AC voltage sensing	1.8 – 5.5v	0.06mA	0.1 – 0.33mW
HCNR200-500E	Precision optocoupler	Isolation with precision analog voltage level balancing	1.6V	10-20mA	16 – 32mW
WS2812B	Addressable LED	Status indicator	5v	0.3mA – 60mA	1.5mW – 300mW
PMOS	mosfet				
NPN	transistor				